

# Golam Murad

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## Research Summary

Machine learning, deep learning, and scientific machine learning (SciML) for civil and environmental engineering, with an emphasis on natural hazards, geospatial modelling, hydrology, remote sensing, uncertainty quantification, and approximate Bayesian learning. Interested in research at the intersection of GeoAI, climate-risk analysis, and physics-informed or uncertainty-aware prediction.

## Education

**Chittagong University of Engineering & Technology (CUET)**  
*B.Sc. in Civil Engineering, CGPA: 3.18/4.00*

July 2025  
Chittagong, Bangladesh

## Research Experience

**Research Engineer**  
*Soil Profile (Private Geotechnical Firm)*

Dec 2025 – Present  
Chittagong, Bangladesh

- Working on landslide and liquefaction assessment for the Chittagong Metropolitan Area.
- Applying geospatial analysis and machine learning workflows for hazard assessment and site characterization.

**Undergraduate Researcher**  
*Department of Civil Engineering, CUET*

2024 – 2025  
Chittagong, Bangladesh

- Conducted thesis research on deep learning-based landslide susceptibility mapping under sparse inventory conditions.
- Worked on explainable AI, Bayesian machine learning, and uncertainty-aware geospatial prediction for natural hazard assessment.

**Industrial Intern**  
*Bangladesh Water Development Board (BWDB)*

Dec 2024  
Chittagong, Bangladesh

## Selected Research Projects

- **Physics-Informed Bayesian Neural Networks for Uncertainty-Aware Landslide Susceptibility Mapping**  
Developed research direction combining domain knowledge, Bayesian deep learning, and geospatial predictors for robust landslide susceptibility assessment under uncertainty.
- **Approximate Bayesian Deep Learning for Inverse Modeling of Ultra-High-Performance Concrete**  
Explored uncertainty-aware inverse design framework for UHPC mixture optimization using approximate Bayesian deep learning.
- **Classical Machine Learning for Flood Susceptibility Mapping with Strict Spatial Holdout**  
Benchmarked eight ML models under spatial holdout with robust preprocessing; Random Forest led ROC-AUC, XGBoost led precision and F1.

## Selected Publications

1. **Murad, G.;** Rahman, M. A.; Yasuhara, H. (2026). “Spatial Sparsity-Aware Explainable Deep Learning-Based Landslide Susceptibility Mapping: Application to a Hill District, Bangladesh.” *EarthArXiv Preprint*. DOI: 10.31223/X52J42. Under review at *Canadian Geotechnical Journal*.
2. **Murad, G.;** Srabony, S. C.; Mahmud, S.; Rahman, M. A. (2026). “Uncertainty-Aware Bayesian Machine Learning for Landslide Susceptibility Mapping in the Chittagong Metropolitan Hill System, Bangladesh.” *EarthArXiv Preprint*. DOI: 10.31223/X55J40.
3. **Murad, G.;** Rahman, M. A.; Das, S. (2026). “Advanced Machine Learning and Explainable AI for Predicting Vegetation Dynamics: A Case Study for Chittagong Division, Bangladesh.” Under review at *Environmental Monitoring and Assessment*.
4. **Murad, G.;** Srabony, S. C.; Gupta, D. D.; Rahman, M. A. (2026). “Stochastic Physics-Informed Neural Network with MC Dropout for Estimating Consolidation Parameters.” Submitted to the *6th Annual Paper Meet of Civil Engineering Division of Institutions of Engineers, Bangladesh (APMCE2026)*.
5. **Murad, G.,** et al. “Landslide Susceptibility Assessment of Rangamati Hill District Using Deep Learning.” Presented at the *8th International Conference on Engineering Research, Innovation and Education*, 2025.
6. **Murad, G.,** et al. “Comprehensive Landslide Inventory and Hotspot Analysis in the Chittagong Hill Tracts of Southeastern Bangladesh.” Presented at the *International Conference on Civil Engineering Research & Innovations*, 2025.

Technical Skills

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|--------------------|---------------------------------------------------------------------|
| Programming        | Python, C, C++, JavaScript                                          |
| ML / Deep Learning | PyTorch, PyTorch Lightning, Scikit-learn, TensorFlow, Keras, Optuna |
| Data Science       | Pandas, NumPy, Matplotlib, Seaborn                                  |
| Geospatial         | ArcGIS Pro, Google Earth Engine, GeoPandas, Rasterio, GDAL          |
| Point Cloud        | Agisoft Metashape                                                   |
| FEM / Structural   | Abaqus, PLAXIS 2D, ETABS, AutoCAD                                   |
| Dev Tools          | Git, GitHub                                                         |

Leadership & Service

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|------------------------------------------------------------------------------|-------------|
| President, ASCE Student Chapter, CUET                                        | 2024 – 2025 |
| President, CUET Debating Society                                             | 2024 – 2025 |
| Volunteer, International Conference on Advances in Civil Engineering (ICACE) | 2022, 2024  |

Certifications

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|-----------------------------------------------------|-------------|
| Landslide Risk Analysis with Remote Sensing and GIS | June 2025   |
| Deep Learning for Earth Observation                 | Aug 2025    |
| Neo4j Knowledge Graphs for RAG                      | In Progress |